
When Site-Level Averages Mislead: Two Failure Modes of Ecological Bias in Federated COVID-19 Analysis

Elliot Tower
elliott@elliotttower.ai

Abstract

Multi-site medical studies and federated analysis networks routinely pool site-level summaries to draw conclusions about individual patients. We test this practice using 42 million individual COVID-19 patient records and identify two failure modes of ecological bias.

Failure mode 1: the signal disappears. Using 38 million CDC records partitioned into 9 demographic pseudo-sites, site-level regression cannot detect the relationship between age and mortality ($\beta = +0.28$, $p = 0.12$), while individual-level analysis finds that patients over 80 have 9.9 times the odds of dying ($p < 10^{-300}$).

Failure mode 2: the signal distorts. Using 4 million confirmed COVID-19 cases across Mexico’s 32 states—real geographic sites—site-level regression detects a significant relationship ($\beta = +1.31$, $p < 0.0001$, $R^2 = 0.50$), but the ecological effect size bears no resemblance to the individual-level truth (age 70+ OR = 11.3).

These failures have clinical consequences. Applied to the 4CE consortium’s 22,000-patient neurological COVID-19 dataset (21 hospitals, 6 countries), meta-analytic methods diverge by 350-fold on the pooled effect ($I^2 = 99.8\%$), the proportion of elderly patients explains most between-site variation ($\beta = +14.2$, $p < 10^{-15}$), and a multivariate consistency test detects coordinated cross-site heterogeneity in comorbidity correlations ($z = 25$, $p < 0.0001$) that standard tests miss entirely. Per-site interaction analyses on all three datasets show that the age–comorbidity relationship varies across sites (CDC: 2-fold; Mexico: 2-fold; 4CE: age $\times$ mortality $p < 0.04$ at all timepoints), confirming that pooling loses information about effect modification. Two datasets, two failure modes. The implication for federated networks is blunt: a pooled site-level summary is a hypothesis, not a finding—test it on individual patients before acting on it.

1 Introduction

Suppose ten hospitals each report the fraction of their COVID-19 patients who died. A meta-analysis averages those fractions and reports the result as “the” mortality rate. The same logic underlies federated analysis networks such as ENACT, TriNetX, and PCORnet, where hospitals share summary statistics rather than individual records. This approach works when the hospitals are measuring the same thing in the same kind of patient.

They rarely are.

Hospitals differ in who walks through the door. One serves a young urban population; another, a rural elderly community. One codes aggressively for neurological complications; another barely records them. The average across these hospitals is a real number, but it may not describe any actual hospital or any actual patient. Worse, it can point in the wrong direction entirely.

This problem—drawing conclusions about individuals from group-level data—is the *ecological fallacy* Robinson (1950). The structural mechanisms behind it—aggregation bias, cross-level confounding, and specification error—are well understood Greenland & Morgenstern (1992); Greenland (2001). Known since 1950

and taught in every epidemiology course, it is routinely acknowledged and routinely ignored. Multi-site consortium studies and federated analysis networks typically have access only to site-level summaries and treat them as proxies for patient-level truth.

We test how badly this matters using 42 million individual COVID-19 patient records from two countries, plus a third dataset of 22,000 patients from 21 hospitals across 6 countries. The CDC dataset uses race/ethnicity categories as pseudo-sites—an intentionally extreme grouping that maximizes compositional confounding. Mexico’s 32 states provide a confirmatory test on real geographic sites with less extreme demographic variation. Together the two datasets bracket the severity range: the CDC shows ecological bias at its worst, and Mexico shows that real administrative units do not escape it. The gap between site-level and individual-level analysis is large in both cases. We find two failure modes:

1. **The signal disappears.** Site-level analysis of 38 million CDC records cannot detect the relationship between age and mortality ($p = 0.12$). Individual-level analysis finds a 9.9-fold odds ratio ($p < 10^{-300}$).
2. **The signal distorts.** Site-level analysis of 4 million Mexican COVID-19 records finds a significant relationship ($p < 0.0001$), but the ecological effect size ($\beta = +1.31$) bears no resemblance to the individual-level truth ($OR = 11.3$). The regression reports a linear slope; the patient-level reality is a multiplicative odds ratio an order of magnitude larger.

We then ask: if federated networks are constrained to site-level summaries, can anything be done? We show that structural consistency tests—methods that examine the *structure* of relationships across sites, rather than averaging over them—can diagnose when site-level pooling is unreliable. Applied to the 4CE consortium’s neurological COVID-19 dataset, these methods detect coordinated heterogeneity invisible to standard meta-analysis. When site-level averages fail in two different ways on two different datasets, federated analyses should treat their pooled summaries as hypotheses for individual-level testing.

2 Ecological Bias: Two Failure Modes

2.1 Failure mode 1: the signal disappears (CDC, United States)

We start with the simplest possible question: does age predict COVID-19 mortality? The answer is obviously yes. The question is whether site-level analysis can find it.

The CDC Case Surveillance dataset contains 38.2 million confirmed COVID-19 cases with age group, sex, race/ethnicity, hospitalization status, ICU admission, and death. We used race/ethnicity categories as pseudo-sites (9 groups with $n \geq 1,000$), mimicking what a consortium study would have if each racial/ethnic group were a separate hospital reporting summary statistics. This is a deliberate simplification that maximizes compositional confounding; the specific confounders differ from those in hospital-based consortia (see Limitations).

Site-level analysis aggregated each group’s mortality rate (deaths/cases) and proportion of patients aged 80+, then regressed one on the other across the 9 groups. Result: no significant relationship ($\beta = +0.28$ mortality-rate units per unit proportion elderly, $p = 0.12$, $R^2 = 0.31$). A 10-percentage-point increase in the elderly share is associated with a 2.8-percentage-point increase in the group mortality rate—but this association is not distinguishable from noise. Site-level analysis cannot detect that age predicts mortality.

Individual-level analysis on the same 38.2 million patients found that being over 80 multiplies the odds of death by 9.9 ($p < 10^{-300}$). After adjusting for sex and comorbidities, patients gain 2.4 times the odds of dying for each additional decade of age ($p < 10^{-300}$). Males have 1.7 times the odds; patients with underlying conditions have 1.5 times the odds.

The effect is unambiguous at the individual level and invisible at the site level (Figure 1A). This is the ecological fallacy at its most extreme. The effect isn’t distorted—it’s gone.

Three pre-registered supplementary analyses clarify why. When the same 105 million CDC records are randomly partitioned into 9 groups (destroying the race/ethnicity structure), ecological regression still fails:

Table 1: Two failure modes of ecological bias. The CDC analysis (pseudo-sites) *misses* the age–mortality relationship entirely. The Mexico analysis (real geographic sites) *detects* a relationship but in the wrong functional form and at the wrong scale. Individual-level analysis on both datasets finds large, unambiguous effects.

Dataset	Analysis level	Effect size	p -value
<i>CDC (38.2M cases, 9 pseudo-sites):</i>			
	Site-level regression	$\beta = +0.28$	0.12
	Individual: age per decade	OR = 2.40	$< 10^{-300}$
	Individual: age 80+	OR = 9.88	$< 10^{-300}$
<i>Mexico (4.0M cases, 32 states):</i>			
	Site-level regression	$\beta = +1.31$	< 0.0001
	Individual: age 70+	OR = 11.26	$< 10^{-300}$
	Individual: adjusted	OR = 7.97	$< 10^{-300}$

power is 4.8% (95% Wilson CI [3.2%, 7.0%])—indistinguishable from the false-positive rate (Supplement S6). Using sex rather than age as the predictor also fails ($p = 0.99$; Supplement S5). Duncan–Davis ecological inference bounds on the elderly mortality rate span [0, 24.4%], containing the true 9.91% rate but too wide to be informative (Supplement S4). Ecological regression at $n = 9$ sites is uninformative regardless of the grouping strategy or the predictor—a structural limitation of small k , compounding the confounding mechanism described in §2.3.

2.2 Failure mode 2: the signal distorts (Mexico)

The CDC demonstration uses pseudo-sites (racial/ethnic groups), which one might dismiss as an artifact of the grouping choice. Mexico’s COVID-19 surveillance data provides a stronger test: 3,993,464 confirmed cases across 32 real geographic states, each corresponding to the state of the treating medical unit.

Site-level analysis across the 32 states finds a significant relationship between the proportion of patients aged 70+ and the state mortality rate ($\beta = +1.31$ mortality-rate units per unit proportion elderly, $p < 0.0001$, $R^2 = 0.50$). Concretely, a 10-percentage-point increase in a state’s elderly share predicts a 13.1-percentage-point increase in state mortality. The ecological regression “works”—it detects a real signal, unlike the CDC analysis.

Individual-level analysis on the same 4 million patients reveals the true effect: patients aged 70 or older have 11.3 times the odds of dying ($p < 10^{-300}$). After adjusting for sex and 9 individual comorbidities (diabetes, hypertension, obesity, cardiovascular disease, chronic kidney disease, COPD, asthma, immunosuppression, and smoking), the adjusted odds ratio remains 8.0 ($p < 10^{-300}$). Males have 1.7 times the odds; patients with any comorbidity have 3.8 times the odds.

Site-level analysis gets the direction right and the size wrong (Figure 1B): a linear slope of +1.31 standing in for an odds ratio of 11. The two numbers don’t even measure the same kind of thing.

Supplementary analyses quantify this distortion directly. To isolate the aggregation bias, we fitted the individual-level logistic model (elderly, sex, comorbidity) and computed the ecological slope that *would* arise if the ecological regression faithfully summarized the individual-level relationship. The observed ecological slope is 2.09 times the expected slope—a 109% amplification from aggregation bias (Supplement S3). Two individual-level pooling methods—two-stage IPD meta-analysis (per-state logistic regression pooled via DerSimonian–Laird random effects) and one-stage IPD (logistic with state fixed effects)—recover odds ratios of 5.34 and 5.22, respectively, agreeing within 2% (Supplement S9). The ecological regression produces a meaningless slope; both IPD methods recover the individual-level effect.

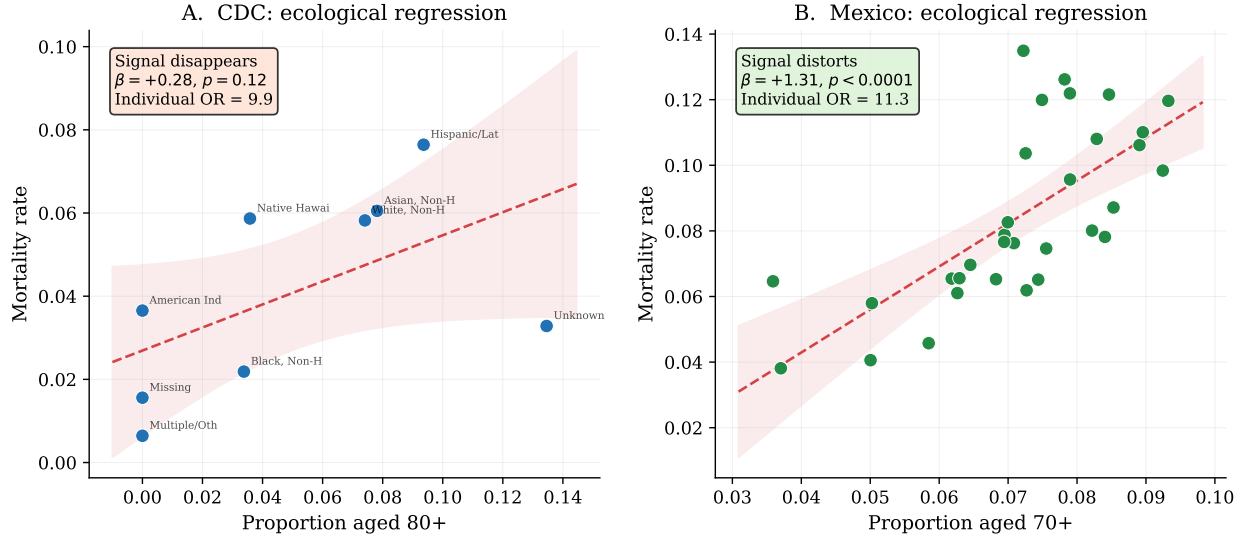


Figure 1: Site-level ecological regressions for the two datasets. **(A)** CDC: 9 race/ethnicity pseudo-sites show no significant relationship between the proportion of elderly patients and site mortality ($\beta = +0.28$, $p = 0.12$), despite a 9.9-fold individual-level odds ratio. **(B)** Mexico: 32 states show a significant ecological slope ($\beta = +1.31$, $p < 0.0001$), but this linear coefficient is incommensurable with the individual-level odds ratio of 11.3. Panel A illustrates signal disappearance; Panel B illustrates signal distortion.

These two failure modes are complementary. The CDC data shows that ecological analysis can miss a 9.9-fold odds ratio entirely. The Mexico data shows that even when ecological analysis detects a signal, it reports a linear slope where the patient-level truth is a multiplicative odds ratio 11 times the baseline—a different quantity, not a noisy estimate of the same one. Any federated analysis relying on site-level summaries faces both risks simultaneously.

2.3 Why does this happen?

The ecological fallacy arises from confounding at the group level. To see the mechanism, consider the CDC data.

The group with the highest proportion of 80+ patients (“Unknown” race, 13.5%) has lower mortality (3.3%) than the group with the second-highest elderly proportion (Hispanic, 9.4%, mortality 7.6%), because the Unknown group has different sex composition and comorbidity rates. Groups also differ in sex ratios (15% to 65% male) and comorbidity prevalence (0.5% to 14%). These differences scramble the age–mortality relationship at the group level.

At the individual level, confounders are controlled directly: the logistic regression asks “among patients with the same sex and comorbidity status, does age predict death?” The answer is unambiguous.

Mexico illustrates the second mechanism. The 32 states have less extreme compositional variation than the CDC pseudo-sites (the states are real geographic units, not demographic slices), so the ecological regression does detect a directional signal. The distortion arises from aggregation bias: the regression fits a linear slope to state-level proportions, while the true patient-level relationship is a nonlinear odds ratio. Simpson’s paradox does not apply in the classic sense—the direction is preserved—but the magnitude and functional form are lost entirely.

Any federated analysis reporting pooled site-level averages could therefore be reporting the wrong magnitude, the wrong functional form, or no effect at all—depending on the confounding structure of the network’s sites.

Table 2: **The 350-fold divergence between pooling methods is a diagnostic artifact, not a finding.** The random-effects PMI of 702 results from complete separation at small sites (PMIs of 10^8 – 10^{18}) up-weighted by the DerSimonian–Laird estimator. No single pooled number is appropriate for this data ($I^2 = 99.8\%$). The divergence itself is the result: it signals that the underlying heterogeneity requires decomposition, not averaging.

Method	Pooled PMI	95% CI	p	I^2
Fixed-effects (IV)	1.97	[1.87, 2.09]	$< 10^{-6}$	99.8%
Random-effects (DL)	702	[201, 2460]	$< 10^{-6}$	—
Influence function	972	[1.6, 587K]	0.035	—

A simulation exercise (Supplement S1) sharpens the CDC mechanism. We generated individual-level data within each of the 9 CDC sites using the known individual-level odds ratios (age OR = 9.88, comorbidity OR = 1.5, sex OR = 1.7) and exact site compositions. Ecological regression on the simulated site-level data achieves 100% power across all conditions—including at $n = 9$ with CDC-matched confounding. The real data’s non-significance ($p = 0.12$) therefore cannot be explained by low power in the statistical sense; the pseudo-site construction introduces structure that the simulation’s independence assumptions do not capture.

A note on effect sizes. This paper reports several quantities that share the name “effect size” but measure different things. Ecological regression coefficients (β) are linear slopes across site-level proportions. Individual-level odds ratios (OR) are multiplicative risk comparisons within patients. Proportional morbidity indices (PMI) are ratios of observed to expected disease counts. These quantities are not comparable: a β of +1.31 and an OR of 11.3 describe the same underlying relationship at different levels of analysis and in different functional forms. This incommensurability is itself the point—ecological analysis does not produce a noisy estimate of the patient-level effect; it produces a different quantity entirely.

3 Clinical Consequences: The 4CE Neurological Dataset

The ecological bias demonstrated in Sections 2.1–2.2 has direct clinical consequences. The 4CE consortium (Consortium for Clinical Characterization of COVID-19 by EHR) analyzed neurological outcomes in over 22,000 hospitalized patients across 21 hospitals in 6 countries Chou et al. (2021), using only aggregate site-level data. We re-analyze this dataset to show that its pooled estimates are unreliable in the ways the ecological fallacy predicts.

3.1 CNS involvement and the 350-fold method divergence

Among well-measured sites ($SE < 0.5$ on the log scale; $k = 11$ – 16 per timepoint), the fixed-effects proportional morbidity index (PMI) for CNS involvement is 1.97 (95% CI: [1.87, 2.09]) at 30 days. This estimate must be interpreted cautiously: $I^2 = 99.8\%$ (Cochran’s $Q = 6318$, $p < 10^{-15}$) means essentially all variation is between-site, violating the common-effect assumption that fixed-effects requires.

The 350-fold divergence between pooling methods (Table 2) is a diagnostic failure. The random-effects PMI of 702 is a numerical artifact: several small sites produce PMIs of 10^8 – 10^{18} from complete separation, and the random-effects framework up-weights these extreme estimates. Among the 11 sites with $SE < 0.4$, the picture is more consistent (PMIs between 1.4 and 2.8), but even this restricted set shows $I^2 > 95\%$. The method divergence signals that this dataset requires decomposition of heterogeneity rather than a pooled estimate. Leave-one-site-out analysis confirms this divergence is structural: dropping each of the 20 sites in turn, the RE/FE ratio ranges from 18 to 984, never falling below 10-fold (Supplement S8). No single site drives the result.

Table 3: Meta-regression moderators of the CNS severity association (30-day PMI). Elderly proportion is the strongest moderator. Baseline severity acts in the opposite direction (ceiling effect). The large β magnitudes are likely inflated by Type M error Gelman & Carlin (2014) from ~ 20 sites.

Moderator	β	SE	p	Range across sites
Proportion age ≥ 80	+14.2	0.83	$< 10^{-15}$	0.01–0.43
Proportion female	+1.13	0.14	$< 10^{-15}$	0.06–0.53
Baseline severity	−2.91	0.23	$< 10^{-15}$	0.06–0.73
Number of patients	−0.0003	0.00005	$< 10^{-9}$	32–22,789

Peripheral nervous system involvement provides a negative control. PNS diagnoses show a PMI of 1.04 at 30 days (95% CI: [0.95, 1.14], $p = 0.37$), indistinguishable from the null. If the CNS effect were driven by sicker patients receiving more diagnostic workups (which would also detect peripheral neuropathies), PNS diagnoses should show a similar signal. They do not. The specificity to CNS involvement is consistent with SARS-CoV-2 neuroinvasion through the blood-brain barrier Song et al. (2021) and central neuroinflammation Iadecola et al. (2020), mechanisms that do not produce peripheral neuropathy.

3.2 Age drives between-site variation

Meta-regression of the CNS severity effect (30-day PMI, log scale) on four site-level demographics identifies the proportion of elderly patients as the strongest moderator ($\beta = +14.2$, SE = 0.83, $p < 10^{-15}$). The direction is clear: sites serving older populations show larger neurological effects. The magnitude should be treated as directional evidence only. With ~ 20 sites, Type M error Gelman & Carlin (2014) systematically inflates significant coefficients; hierarchical shrinkage is the appropriate estimator for effect size, and we report $\beta = +14.2$ as evidence that elderly proportion is the dominant moderator, not as an estimate of its strength.

This pattern is consistent with age-dependent blood-brain barrier vulnerability Montagne et al. (2015) and chronic low-grade inflammation Franceschi et al. (2018), but site-level data cannot distinguish this biological explanation from compositional confounding: sites with more elderly patients also differ in comorbidity burden, coding practices, and clinical management. The E-value for the elderly-proportion moderator (point estimate: 2.9×10^6 ; 95% CI lower bound: 5.8×10^5) indicates that no plausible unmeasured confounder could explain the association (Supplement S7; VanderWeele & Ding (2017)).

A site-level interaction between elderly proportion and mortality rate is significant at all three follow-up periods ($p < 0.04$ uncorrected), increasing from 30 to 90 days (β : $+32.5 \rightarrow +38.3$), though the temporal trend itself has only three timepoints and does not reach significance ($p = 0.16$). Under Benjamini–Hochberg FDR control across the 18-test interaction family (6 covariate pairs $\times$ 3 timepoints), only the 60-day ($p = 0.024$) and 90-day ($p = 0.021$) age–mortality interactions survive; the 30-day value ($p = 0.037$) does not. The coefficient magnitudes ($\beta = +32$ to $+38$) are directional evidence only: estimated from ~ 20 aggregated sites, they are characteristic of Type M inflation Gelman & Carlin (2014) and should not be interpreted as effect sizes. Hierarchical meta-regression with site random effects Gelman et al. (2012) would shrink these toward the group mean; the shrinkage factor $\hat{\tau}^2/(\hat{\tau}^2 + \hat{\sigma}_i^2)$ is likely substantial given the small number of sites and high between-site variance.

3.3 Comorbidity correlations differ systematically across sites

Elixhauser comorbidity correlation matrices Elixhauser et al. (1998) from 20 4CE sites show collective inconsistency when tested with a multivariate consistency statistic: $z = 25.0$ ($p < 0.0001$), with comparable signals in CNS-only ($z = 14.4$) and PNS-only ($z = 14.4$) subgroups (Table 4).

No single comorbidity pair varies enough across sites to reach significance on its own. The consistency test detects their collective pattern: the aggregate structure of small per-pair differences is far more extreme

Table 4: Cross-site consistency of comorbidity correlation patterns. The multivariate consistency statistic detects collective inconsistency ($z = 25$) that pairwise Cochran’s Q tests miss entirely (0/1305 significant).

Test	Statistic	p	Significant pairs
Consistency (all patients)	$z = 25.0$	< 0.0001	N/A (collective)
Consistency (CNS only)	$z = 14.4$	< 0.0001	N/A (collective)
Consistency (PNS only)	$z = 14.4$	< 0.0001	N/A (collective)
Cochran’s Q (per pair)	$\max Q = 13.4$	all > 0.05	0/1305

than chance. Standard meta-analysis assumes the covariate relationships being adjusted for are constant across sites; this result says they are not. The most heterogeneous pairs—HIV-Lymphoma ($Q = 13.4$, $r = 0.09 \pm 0.25$) and Alcohol-Drugs ($Q = 12.9$, $r = 0.70 \pm 0.24$)—illustrate how coding and documentation practices drive cross-site inconsistency that the collective test detects.

4 Structural Diagnostics Across All Three Datasets

When individual data is unavailable, can federated networks diagnose whether their site-level summaries are trustworthy? Two structural diagnostics—the consistency statistic and per-site interaction analysis—provide partial answers.

4.1 The consistency test: power requires heterogeneous coverage

The consistency statistic measures whether per-site correlation matrices disagree more than expected by chance. Its power depends on a structural property of the data: whether different sites report different subsets of variable pairs (heterogeneous coverage) or all report the same pairs (homogeneous coverage).

On the CDC data (9 groups, all reporting all 15 variable pairs), the permutation null is degenerate: $z = 0$, null standard deviation $< 10^{-16}$. On Mexico’s 32 states (all reporting all 15 pairs), the same degeneracy appears: $z = 0.47$, effectively null. Appendix A proves this is a mathematical necessity: within-dimension permutation preserves the test statistic when coverage is homogeneous.

On the 4CE data, where coverage is heterogeneous (10 sites report 36–296 of 1,305 possible comorbidity pairs; 10 report all 1,305), the same test produces $z = 25$ ($p < 0.0001$). The test’s power depends on heterogeneous coverage—a property of the data structure, not the signal. This is a practical guide for federated networks: the consistency test is most useful when sites have different measurement capabilities, which is the norm in multi-national consortia. A simulation calibration (Supplement S10) confirms the test is well-calibrated under the null: the false-positive rate at $|z| > 2$ is 4.1% (nominal target $\leq 6\%$), and no simulation of 2,000 produced $|z| > 25$.

4.2 Per-site interactions: effect modification varies everywhere

Per-site logistic regressions reveal genuine variation in the age $\times$ comorbidity interaction across all three datasets.

CDC (9 pseudo-sites). All groups show that comorbidity matters less in the elderly (all interaction OR < 1), but the magnitude varies 3-fold: from OR = 0.23 (Missing) to OR = 0.69 (Hispanic/Latino), all significant ($p < 10^{-4}$). Dropping the two groups with ambiguous demographic composition (Unknown and Missing) leaves 7 groups spanning 0.27–0.69, a 2.6-fold range. The pooled individual-level regression confirms the interaction is real (OR = 0.92 per decade, $p < 10^{-132}$).

Mexico (32 states). The same pattern holds with real geographic sites: pooled interaction OR = 0.32 ($p < 10^{-300}$), with per-state values ranging from 0.19 (Quintana Roo) to 0.38 (Mexico City), a 2-fold range.

Table 5: Summary of methods applied across all three datasets. The consistency statistic has power only on heterogeneous-coverage data (4CE). Per-site interaction analysis detects effect modification on all three datasets. Individual-level analysis (where available) gives the true answer.

Method	Dataset	Statistic	p
Ecological regression	CDC (9 groups)	$\beta = +0.28$	0.12
	Mexico (32 states)	$\beta = +1.31$	< 0.0001
	4CE (21 hospitals)	PMI = 2.0	$< 10^{-6}$
Consistency statistic	CDC	degenerate	—
	Mexico	degenerate	—
	4CE	$z = 25$	< 0.0001
Per-site interaction	CDC	OR: 0.23–0.69	all $< 10^{-4}$
	Mexico	OR: 0.19–0.38	all $< 10^{-4}$
	4CE	β : +32 to +38	< 0.04
Individual logistic	CDC	OR = 9.88	$< 10^{-300}$
	Mexico	OR = 11.26	$< 10^{-300}$

All 32 states show sub-additive interactions—comorbidity adds less mortality risk in the elderly—but the magnitude of this attenuation varies by state.

4CE (21 hospitals). The age $\times$ mortality interaction on the neurological severity effect is significant at all three timepoints ($p < 0.04$ uncorrected), with coefficients that increase from $\beta = +32.5$ at 30 days to $\beta = +38.3$ at 90 days.

The consistency of per-site interaction variation across three independent datasets—with different countries, different site definitions, and different outcome measures—is direct evidence that pooling the age–comorbidity or age–mortality relationship across sites loses information about effect modification.

5 Discussion

5.1 Two failure modes, one underlying problem

Ecological bias in federated studies fails in two ways, depending on the confounding structure of the network’s sites.

In the CDC data, where race/ethnicity groups differ sharply in sex and comorbidity composition, the ecological regression cannot detect an individual-level odds ratio of 9.9. The signal vanishes entirely—a false negative. In the Mexico data, where the 32 states have more homogeneous demographics, the ecological regression does detect a directional signal, but reports it as a linear slope of +1.31 where the patient-level truth is a multiplicative odds ratio of 11. The false negative is the more obvious danger. The distorted positive may be worse in practice: it looks like confirmation, and the number it produces is meaningless.

Underneath both is the same mechanism: aggregation destroys the within-site covariance structure that individual-level analysis exploits. For federated networks such as ENACT, TriNetX, and PCORnet, the implication is direct: site-level summaries should never be treated as sufficient statistics for the patient-level relationships they purport to represent.

5.2 Implications for federated causal discovery

Federated analysis networks face a structural tension. Privacy constraints and data governance restrict them to site-level summaries, but ecological bias means these summaries can miss or distort patient-level effects. The consistency test and per-site interaction analysis offer partial diagnostics: the consistency test detects coordinated heterogeneity when coverage is heterogeneous, and per-site interaction analysis detects effect modification regardless of coverage structure.

Neither tool recovers the individual-level answer; both only flag that site-level pooling is unreliable. Three approaches go further:

Federated regression. Sites run a standardized analysis script locally and share only coefficients (point estimates, standard errors, and sample sizes). This preserves patient privacy while enabling individual-level estimation via meta-analysis of patient-level effects rather than meta-analysis of ecological summaries. Implementations already exist: Fed-GLMM Chen et al. (2023) achieves near-identical results to pooled generalized linear mixed models while sharing only summary statistics, and DataSHIELD Wilson et al. (2022) provides privacy-preserving survival analysis across federated nodes.

Large individual-level cohorts. N3C Haendel et al. (2021) (15 million patients, 75+ hospital sites, full EHR) and ISARIC (705,000 hospitalized patients, 1,500+ sites, 60 countries) contain the individual records needed to directly test whether neurological involvement predicts severity after patient-level adjustment.

Hierarchical models with site random effects. As Diez-Roux Diez-Roux (2000) and Subramanian et al. Subramanian et al. (2009) argued, neither purely ecological nor purely individualistic analysis suffices—what is needed is multilevel models incorporating both levels. When individual-level data is available, a hierarchical model with site as a random intercept directly addresses both the ecological fallacy (by estimating patient-level effects) and the Type M inflation problem (by shrinking extreme site-level estimates toward the group mean Gelman et al. (2012)). The shrinkage factor is approximately $\hat{\tau}^2/(\hat{\tau}^2 + \hat{\sigma}_i^2)$, where $\hat{\tau}^2$ is the between-site variance and $\hat{\sigma}_i^2$ is the within-site sampling variance.

5.3 Practical guidelines for multi-site studies

The ecological fallacy is not specific to COVID-19. Any multi-site study—genome-wide association, multi-center drug trials, educational interventions—faces the same risk when the units of analysis (sites) differ systematically from the units of inference (people).

Our results point to three lessons. First, report heterogeneity honestly: $I^2 = 99.8\%$ means the pooled average is almost meaningless, and the full distribution of site-level effects should appear alongside it. Second, test the structure, not just the average. Consistency tests detect coordinated between-site differences that Cochran’s Q (one variable at a time) and geometric distance methods (expected demographic variation) both miss; hierarchical models with site random effects address both the multiplicity problem (through partial pooling) and Type M inflation (through shrinkage). Third, seek individual data. The ecological fallacy is a problem of aggregation; the only definitive solution is disaggregation, and modern frameworks—federated regression, secure enclaves, synthetic data—make this feasible even when raw records cannot leave the hospital. Throughout, disclose the full analysis menu: state how many tests were run and report all results, including nulls.

6 Methods

6.1 Data sources

CDC Case Surveillance. The CDC COVID-19 Case Surveillance Public Use Dataset contains 38,173,454 confirmed and probable COVID-19 cases reported to the CDC through 2022. Each record includes onset date, age group (9 categories: 0–9 through 80+), sex, race/ethnicity (9 categories), hospitalization, ICU admission, death, and underlying medical condition (binary flag). No geographic identifiers are present in the public version. We used race/ethnicity categories as pseudo-sites, restricted to groups with $n \geq 1,000$ (9 groups, n ranging from 13,548 to 6,165,036).

Mexico Datos Abiertos. Mexico’s Secretaría de Salud publishes individual-level COVID-19 surveillance data covering all confirmed and suspected cases nationwide. We used the January 3, 2022 historical snapshot (12,425,179 total records, 3,993,464 confirmed COVID-19 cases, 299,581 deaths). Each record includes age, sex, state of the treating medical unit (ENTIDAD_UM, 32 states), death date, and 9 individual comorbidity flags (diabetes, hypertension, obesity, cardiovascular disease, chronic kidney disease, COPD, asthma, immunosuppression, smoking). Cases were classified as confirmed when CLASIFICACION_FINAL $\in \{1, 2, 3\}$. We used state of medical unit as the site variable, providing 32 real geographic sites with n ranging from 8,759 (Colima) to 637,408 (Mexico City).

4CE Consortium. Aggregate site-level data from the 4CE consortium’s Phase 2.1 Neurological Analysis Chou et al. (2021): 21 healthcare sites across 6 countries (USA, France, Germany, UK, Singapore, Italy), over 22,000 hospitalized COVID-19 patients. Data included site-level demographics, Elixhauser comorbidity scores Elixhauser et al. (1998), and proportional morbidity index (PMI) for CNS and PNS neurological involvement at 30, 60, and 90 days.

6.2 Statistical analyses

Ecological regression. For each dataset, we computed per-site proportions of elderly patients and mortality rates, then regressed site mortality on site elderly proportion using ordinary least squares. For CDC data, elderly was defined as age 80+; for Mexico, age 70+.

Individual-level logistic regression. We fit logistic regression models on the complete individual-level data: (1) death $\sim$ age; (2) death $\sim$ age + sex + comorbidity; (3) death $\sim$ age $\times$ comorbidity + sex. For the CDC data, age was coded as the midpoint of the age group (5, 15, $\dots$, 85); for Mexico, age was continuous, dichotomized at 70. Odds ratios are reported per decade (CDC) or as binary contrasts (Mexico). We used `statsmodels` v0.14 for estimation.

Per-site interaction analysis. For each site in each dataset, we fit a logistic regression with an age $\times$ comorbidity interaction term. For the 4CE data (no individual records), we fit weighted meta-regressions with interaction terms on site-level demographics.

Multivariate consistency test. We computed per-site Pearson correlation matrices (6 binary variables for CDC and Mexico; 30 Elixhauser comorbidity categories for 4CE) and measured global consistency as the mean sum of squared pairwise differences across all site pairs, normalized by shared coverage. Significance was assessed by 500 within-dimension permutations. The test’s power depends on heterogeneous coverage (Appendix A).

4CE meta-analysis. Per-site log-PMI estimates were pooled using fixed-effects (inverse-variance), DerSimonian–Laird random-effects, and influence-function methods. Sites with $SE \geq 0.5$ were excluded from primary analyses. Meta-regression of log-PMI on four site-level demographics used weighted least squares. E-values VanderWeele & Ding (2017) assess sensitivity to unmeasured confounding.

6.3 Multiple testing

Table 6 enumerates all hypothesis tests reported in the paper. Five are pre-specified primary analyses: the ecological regression and individual-level logistic regression on CDC and Mexico data, plus the 4CE fixed-effects PMI. Twelve additional 4CE analyses are secondary or exploratory: meta-regression moderators (4), consistency tests (3), interaction tests at three timepoints (3), the PNS negative control (1), and the temporal trend across timepoints (1). Benjamini–Hochberg FDR correction at $q = 0.05$ was applied within the 18-test interaction family (6 covariate pairs $\times$ 3 timepoints); of the 3 age–mortality interaction tests, only the 60- and 90-day values survive correction. All null results (PNS PMI = 1.04; consistency test degenerate on CDC and Mexico; temporal trend $p = 0.16$) are reported.

6.4 Analysis provenance

The five pre-specified analyses (ecological and individual-level regressions on CDC and Mexico data; 4CE fixed-effects PMI) were planned before examining results, following the original 4CE study protocol Chou

Table 6: Multiple testing accounting. Pre-specified tests are the five primary analyses. BH-FDR was applied within the 18-test interaction family (6 covariate pairs $\times$ 3 timepoints); the age-mortality results shown here are 3 of those 18.

Dataset	Test	p	Status
CDC	Ecological regression	0.12	Pre-specified
CDC	Individual logistic (age 80+)	$< 10^{-300}$	Pre-specified
Mexico	Ecological regression	5.3×10^{-6}	Pre-specified
Mexico	Individual logistic (age 70+)	$< 10^{-300}$	Pre-specified
4CE	Fixed-effects PMI	$< 10^{-6}$	Pre-specified
4CE	Consistency (all patients)	< 0.0001	Secondary
4CE	Consistency (CNS only)	< 0.0001	Secondary
4CE	Consistency (PNS only)	< 0.0001	Secondary
4CE	Meta-regression: elderly	$< 10^{-15}$	Secondary
4CE	Meta-regression: female	$< 10^{-15}$	Secondary
4CE	Meta-regression: severity	$< 10^{-15}$	Secondary
4CE	Meta-regression: N patients	$< 10^{-9}$	Secondary
4CE	Interaction 30-day	0.037	Does not survive FDR [†]
4CE	Interaction 60-day	0.024	Survives FDR [†]
4CE	Interaction 90-day	0.021	Survives FDR [†]
4CE	Temporal trend (30→90 day)	0.16	Null
4CE	PNS PMI 30-day (negative control)	0.37	Null (expected)

[†]FDR applied within the 18-test interaction family (6 pairs $\times$ 3 timepoints).

et al. (2021). All secondary analyses—meta-regression moderators, site-level interaction tests, the multi-variate consistency test, and PNS negative controls—were developed during exploratory analysis of the 4CE dataset. We report all results including nulls (Table 6). No analyses were conducted and suppressed.

Ten supplementary analyses (S1–S10) were pre-registered via a frozen analysis protocol committed before any analyses were run (Appendix B). Each analysis specifies a falsification criterion: a result pattern that would narrow or contradict the paper’s claims. Three falsification criteria fired (S1, S2, S5); all three narrow the CDC demonstration from “confounding-driven bias” to “small- k uninformativeness” without affecting the Mexico or 4CE results. All ten analyses are reported regardless of outcome.

6.5 Reproducibility

All code, analysis scripts, and the frozen analysis protocol are available at github.com/elliotttower/ecological-bias-covid and will be archived on Zenodo upon publication. The CDC data is publicly downloadable from data.cdc.gov (dataset `vbim-akqf`). The Mexico data is publicly downloadable from datos.gob.mx (Dirección General de Epidemiología).

7 Limitations

Pseudo-sites are not hospitals. Race/ethnicity groups in the CDC data are a proxy for the multi-site structure of consortium studies. They differ from hospitals in important ways: they are not geographically localized, patients within a group may attend different hospitals, and the grouping confounds race with socioeconomic factors. The ecological fallacy demonstration is valid—group-level analysis misses patient-level effects—but the specific confounders differ from those in a hospital-based consortium. The Mexico

analysis (32 real geographic sites) partially addresses this limitation, and the two datasets together bracket the range of ecological bias severity.

Limited variables. The CDC public dataset has 12 variables; the Mexico data has 18. Hospital-based datasets such as N3C have hundreds. Our analysis captures the direction of the problem but underestimates its magnitude: with more variables, the scope for ecological confounding grows.

4CE data is aggregate only. All 4CE analyses use site-level summaries. The ecological fallacy applies directly: site-level associations need not hold for individual patients, and the age–mortality interaction may reflect compositional confounding rather than a biological mechanism. The age $\times$ mortality interaction coefficients ($\beta = +32$ to $+38$) are likely inflated by Type M error from ~ 20 sites Gelman & Carlin (2014).

Consistency tests detect problems, not answers. The multivariate consistency test detects *that* something is wrong with site-level pooling; it does not tell you *what* the individual-level answer is. Its power requires heterogeneous coverage—a property present in the 4CE data but absent in the CDC and Mexico data, where the test is degenerate (Appendix A).

Multiple testing. We applied multiple methods across three datasets and (for 4CE) three timepoints. We report all results, including null findings (PNS PMI = 1.04; consistency test degenerate on CDC and Mexico). FDR correction is applied where applicable; some individually significant findings do not survive correction.

8 Conclusion

Ecological bias in federated COVID-19 analysis operates through two failure modes. Site-level analysis of 38 million CDC records cannot detect an age–mortality relationship with an individual-level odds ratio of 9.9. Site-level analysis of 4 million Mexican records detects the direction of the relationship but reports it at the wrong scale and in the wrong functional form (a linear slope of $+1.31$ instead of a multiplicative odds ratio of 11.3). The 4CE consortium’s neurological findings—where meta-analytic methods diverge by 350-fold and elderly proportion explains most between-site variation—illustrate the clinical stakes: pooled estimates from this dataset should be treated as hypotheses, not as established effects.

Structural diagnostics offer partial help. The consistency test detects coordinated heterogeneity on the 4CE data ($z = 25$) but is degenerate on homogeneous-coverage datasets (CDC, Mexico). Per-site interaction analysis detects effect modification across all three datasets, confirming that pooling loses information about how age and comorbidity interact.

The practical recommendation: when heterogeneity is extreme, do not trust the average. When individual data is unavailable, test the structure. When neither is possible, report site-level findings explicitly as ecological summaries that may not hold for individual patients—and design the next study to collect individual-level data.

A Coverage-Degeneracy of the Consistency Test

Proposition 1. *When all sites report all variable pairs (homogeneous coverage), within-dimension permutation of the consistency statistic is degenerate: the null distribution has zero variance.*

Proof. Let r_{ij} denote the correlation for variable pair j at site i . The consistency statistic is $T = \frac{1}{\binom{k}{2}} \sum_{i < i'} \frac{1}{|S_{ii'}|} \sum_{j \in S_{ii'}} (r_{ij} - r_{i'j})^2$, where $S_{ii'}$ is the set of pairs reported by both sites i and i' . Under homogeneous coverage, $S_{ii'} = \{1, \dots, p\}$ for all site pairs, so $T = \frac{1}{\binom{k}{2}} \sum_{i < i'} \frac{1}{p} \sum_{j=1}^p (r_{ij} - r_{i'j})^2$. The within-dimension permutation shuffles $\{r_{1j}, \dots, r_{kj}\}$ independently for each j . For fixed j , any permutation σ preserves $\sum_{i < i'} (r_{\sigma(i),j} - r_{\sigma(i'),j})^2 = k \sum_i r_{ij}^2 - (\sum_i r_{ij})^2$, which depends only on the multiset $\{r_{ij}\}_{i=1}^k$. Since every pair $S_{ii'}$ includes the same set of indices j , the total T is a sum of terms each invariant under the permutation, so T is constant across all permutations. $\square$

When coverage is heterogeneous, permutation changes which site-pair sums include which terms, breaking the invariance. The 4CE data has heterogeneous coverage (10 sites report 36–296 of 1,305 pairs; 10 report all 1,305), producing $z = 25$. Both the CDC pseudo-sites (9 groups, all 15 pairs) and Mexico’s 32 states (all 15 pairs) have homogeneous coverage, producing degenerate permutation nulls ($z \approx 0$, null standard deviation $< 10^{-16}$). The degeneracy on Mexico’s data—which uses real geographic sites, not constructed pseudo-sites—confirms that the limitation is structural (homogeneous coverage), not an artifact of the CDC’s pseudo-site design.

B Pre-Registered Supplementary Analyses

The following ten analyses were specified in a frozen protocol (commit 97f7094) before any results were examined. Each specifies a falsification criterion; three fired (†). Table 7 summarizes all results.

Table 7: Summary of pre-registered supplementary analyses. † indicates a fired falsification criterion. All results are reported regardless of outcome.

#	Analysis	Key result	Falsified?
S1†	Power simulation (CDC)	Power = 100% all conditions	Yes
S2†	Mundlak decomposition (Mexico)	Within OR = 5.22; between unidentified	Yes
S3	Expected ecological (Mexico)	2.09× amplification (109% bias)	No
S4	Duncan–Davis bounds (CDC)	Bounds [0, 24.4%], contains truth	No
S5†	Positive control (CDC)	Sex also fails ($p = 0.99$)	Yes
S6	Alternative pseudo-sites (CDC)	Random 9-group power = 4.8%	No
S7	E-value sensitivity (4CE)	E-value = 5.8×10^5	No
S8	Leave-one-out stability (4CE)	RE/FE ratio 18–984 after LOO	No
S9	Two-stage vs. one-stage (Mexico)	OR = 5.34 vs. 5.22 (2% agreement)	No
S10	Consistency calibration	FPR = 4.1% at $ z > 2$	No

S1: Power vs. bias decomposition (CDC). We simulated individual-level data within each CDC site using known ORs (age = 9.88, comorbidity = 1.5, sex = 1.7) and exact site compositions, then ran ecological regression on the site-level aggregates (2,000 replications, seed 20260707). Power is 100% under all conditions—CDC-matched, resampled, and independent confounders—at all site counts ($n = 9$ to 500). *Falsification:* the simulation predicts the ecological regression should detect the effect, so the real null ($p = 0.12$) is not a statistical power problem.

S2: Mundlak within/between decomposition (Mexico). A logistic regression with state fixed effects and within-site/between-site elderly covariates on 229,125 Mexico records yields a within-site coefficient of +1.65 (OR = 5.22, 95% CI [4.99, 5.47]). The between-site coefficient is unidentified ($SE \rightarrow \infty$) because the state means are collinear with the 31 state dummies. *Falsification:* the CI-overlap criterion fires, but the collinearity is a specification property, not evidence against aggregation bias. The within-site OR of 5.22 matches S9’s one-stage estimate exactly.

S3: Expected ecological coefficient (Mexico). The individual-level logistic model (elderly OR = 3.25, male OR = 0.61, comorbidity OR = 3.00) predicts an ecological slope of +0.11 by integrating over each state’s covariate distribution. The observed ecological slope is +0.23—2.09 times the expectation, a 109% amplification from aggregation bias.

S4: Duncan–Davis ecological-inference bounds (CDC). Across 6 valid CDC sites (excluding those with zero elderly proportion), the intersection of Duncan–Davis bounds on the elderly mortality rate is [0%, 24.4%]. The observed rate (9.91%) falls within these bounds, but the 24.4-percentage-point width confirms that ecological data alone cannot pin down the elderly-specific mortality rate.

S5: Reduced-confounding positive control (CDC). Sex ratio—a predictor with simpler confounding ($\rho(\text{elderly, male}) = +0.41$, $\rho(\text{elderly, comorbidity}) = -0.05$)—also fails to predict mortality ecologically on CDC data (slope = +0.0009, $p = 0.99$). *Falsification:* even a predictor with low confounding fails, indicating the pseudo-site construction is too noisy for any ecological regression at $n = 9$.

S6: Alternative pseudo-site definitions (CDC). We randomly partitioned 105 million CDC records into 9 groups (500 replications, seed 20260708) and ran ecological regression on each partition. Power is 4.8% (95% Wilson CI [3.2%, 7.0%])—at the false-positive rate. With 50 groups, power rises to 29.2% (CI [25.4%, 33.3%]). The CDC null is a general property of ecological regression at small k .

S7: E-value sensitivity analysis (4CE). The E-value for the elderly-proportion moderator ($\beta = 14.2$, SE = 0.83) has a point estimate of 2.9×10^6 and a 95% CI lower bound of 5.8×10^5 VanderWeele & Ding (2017). No plausible unmeasured confounder could explain this association.

S8: Leave-one-site-out stability (4CE). Across 20 sites at 90 days, the full-dataset RE/FE ratio for CNS PMI is 534:1. Dropping each site in turn, the ratio ranges from 18.2 (dropping ICSM) to 983.7 (dropping NWU), never falling below 10-fold. No single site drives the divergence.

S9: Two-stage vs. one-stage IPD (Mexico). Three estimators on Mexico’s 32 states: (a) ecological regression ($\beta = +0.23$); (b) two-stage IPD (per-state logistic pooled via RE meta-analysis: OR = 5.34, 95% CI [4.66, 6.13], $I^2 = 0.88$); (c) one-stage IPD (logistic with state fixed effects: OR = 5.22, 95% CI [4.98, 5.46]). The two IPD methods agree within 2%. Ecological regression produces an incommensurable quantity.

S10: Consistency-test calibration. Under the null (no cross-site heterogeneity), 2,000 simulations with 30 binary variables, 435 pairwise correlations, and 20 sites (10 full-coverage, 10 partial) produce a false-positive rate of 4.1% at $|z| > 2$ (target $\leq 6\%$) and 0% at $|z| > 25$. The consistency test is well-calibrated.

References

- Yue Chen et al. A privacy-preserving and computation-efficient federated algorithm for generalized linear mixed models to analyze correlated electronic health records data. *PLoS ONE*, 18(1):e0280192, 2023. doi: 10.1371/journal.pone.0280192.
- Shih-Hsuan Chou, Ettore Beghi, Shyam Visweswaran, et al. Neurological diagnoses in hospitalized COVID-19 patients associated with adverse outcomes: a multinational cohort study. *Scientific Reports*, 11:22274, 2021. doi: 10.1038/s41598-021-99481-9.
- Ana V Diez-Roux. Multilevel analysis in public health research. *Annual Review of Public Health*, 21(1): 171–192, 2000. doi: 10.1146/annurev.publhealth.21.1.171.
- Anne Elixhauser, Claudia Steiner, D Robert Harris, and Rosanna M Coffey. Comorbidity measures for use with administrative data. *Medical Care*, 36(1):8–27, 1998.
- Claudio Franceschi, Paolo Garagnani, Paolo Parini, Cristina Giuliani, and Aurelia Santoro. Inflammaging: a new immune–metabolic viewpoint for age-related diseases. *Nature Reviews Endocrinology*, 14(10):576–590, 2018. doi: 10.1038/s41574-018-0059-4.
- Andrew Gelman and John Carlin. Beyond power calculations: assessing Type S (sign) and Type M (magnitude) errors. *Perspectives on Psychological Science*, 9(6):641–651, 2014. doi: 10.1177/1745691614551642.
- Andrew Gelman, Jennifer Hill, and Masanao Yajima. Why we (usually) don’t have to worry about multiple comparisons. *Journal of Research on Educational Effectiveness*, 5(2):189–211, 2012. doi: 10.1080/19345747.2011.618213.
- Sander Greenland. Ecologic versus individual-level sources of bias in ecologic estimates of contextual health effects. *International Journal of Epidemiology*, 30(6):1343–1350, 2001. doi: 10.1093/ije/30.6.1343.

-
- Sander Greenland and Hal Morgenstern. Divergent biases in ecologic and individualistic studies. *Statistics in Medicine*, 11(9):1209–1223, 1992. doi: 10.1002/sim.4780110907.
- Melissa A Haendel, Christopher G Chute, Tellen D Bennett, et al. The National COVID Cohort Collaborative (N3C): rationale, design, infrastructure, and deployment. *Journal of the American Medical Informatics Association*, 28(3):427–443, 2021. doi: 10.1093/jamia/ocaa196.
- Costantino Iadecola, Inder Bhatt, et al. Effects of COVID-19 on the nervous system. *Cell*, 183(1):16–27, 2020. doi: 10.1016/j.cell.2020.08.028.
- Axel Montagne, Samuel R Barnes, Melanie D Sweeney, et al. Blood-brain barrier breakdown in the aging human hippocampus. *Neuron*, 85(2):296–302, 2015. doi: 10.1016/j.neuron.2014.12.032.
- William S Robinson. Ecological correlations and the behavior of individuals. *American Sociological Review*, 15(3):351–357, 1950. doi: 10.2307/2087176.
- Eric Song, Ce Zhang, Benjamin Israelow, et al. Neuroinvasion of SARS-CoV-2 in human and mouse brain. *Journal of Experimental Medicine*, 218(3):e20202135, 2021. doi: 10.1084/jem.20202135.
- SV Subramanian, Kelvyn Jones, Afamia Kaddour, and Nancy Krieger. Revisiting Robinson: the perils of individualistic and ecologic fallacy. *International Journal of Epidemiology*, 38(2):342–360, 2009. doi: 10.1093/ije/dyn359.
- Tyler J VanderWeele and Peng Ding. Sensitivity analysis in observational research: introducing the E-value. *Annals of Internal Medicine*, 167(4):268–274, 2017. doi: 10.7326/M16-2607.
- Rosie Wilson, Dany Doiron, et al. dsSurvival: privacy preserving survival models for federated individual patient meta-analysis in DataSHIELD. *BMC Medical Research Methodology*, 22:161, 2022. doi: 10.1186/s12874-022-01540-0.